

6 V = 12 V / 2, buffered, so every stage shares one quiet reference.

f = $V_{sw} / (8 R C)$: $R4 + R5 = 5k\Omega$ with $C7 = 470pF$ gives $250 kHz$; $R5/R6$ sets the $\pm 2 V$ window around $0 V$

$f = V_{sw} / (8 R C)$: $R_4 + R_{V1} \sim 5k\Omega$ with $C_{12} 470p$ gives 250 kHz; R_5/R_6 set the ± 2 V window about 6 V

Then the circuit is as follows:

Test points TP1...TP5 are solder lugs on TRI, PWM_A, both switching nodes and the filtered output.

Bring up with NO MOSFETs fitted: check 12 V and 6 V, then the triangle, then both PWM outputs, then the HIP4082 dead time on a scope. Only then fit Q1,Q4, on a current-limited supply into a dummy load.

The image shows three circuit diagrams for generating square waves using four CD4049UBE inverters (U5A, U5B, U5C, U5D). Each inverter has its non-inverting input connected to ground. The outputs of U5A and U5B are connected to the inputs of U5C and U5D, respectively. The outputs of U5C and U5D are labeled A_HI and B_HI. The outputs of U5A and U5B are labeled A_LO and B_LO. The outputs of U5C and U5D are labeled A_HI and B_HI. The outputs of U5A and U5B are labeled A_LO and B_LO. The outputs of U5C and U5D are labeled A_HI and B_HI.

[illegible]

The diagram shows a 4-channel relay module with two 5V relays (Q1, Q2) and two 12V relays (Q3, Q4). The module is powered by a +5V supply and a +12V supply. The ground is labeled GND. The 5V relays are controlled by signals G_AH, G_AL, G_BH, and G_BL through 22kΩ resistors (R14, R15, R16, R17). The 12V relays are controlled by signals SW_A and SW_B through 10kΩ resistors (R18, R19, R20, R21). The module has two output terminals, TP3 and TP4, which are connected to the common terminals of the relays. The module is labeled with '540N' and '12V'.

The schematic diagram illustrates the output stage of the amplifier. The collector of the transistor is connected to a load resistor R22 (10k 5W) and a bypass capacitor F14 (100n). The emitter is connected to ground through a resistor R13 (0.20n) and a capacitor C4 (2200uF). The output is taken from the collector through a coupling capacitor F13 (100n) to a speaker (SPEAKER 4R). The input is connected to TP5 and OUT_P.

- A rails and the 6 V virtual ground → `sim/sim_a_vground.kicad_pro`
- B triangle carrier, -250 kHz → `sim/sim_b_triangle.kicad_pro`
- C input stage, buffer and inverter → `sim/sim_c_input.kicad_pro`
- D PWM comparators → `sim/sim_d_pwm.kicad_pro`